

Agentic Zealandia: A Population-Scale Human–AI Coordination Infrastructure

Core Idea: Agentic Zealandia proposes building a population-scale human–AI coordination infrastructure for a nation. The goal is not to replace people with AI, nor to create a centralized surveillance system, but to model, test, and deploy coordination protocols that help humans and AI systems work together effectively across critical domains such as work, health, and the environment. The system focuses on how millions of decisions interact, where coordination failures create cascading risks, and how those risks can be anticipated and mitigated before they become crises.

Overview: We propose a partnership between PHF Science and MIT Project Iceberg to build Agentic Zealandia, a national coordination layer that connects population-scale simulation with real-world deployment. The platform integrates three capabilities: simulating how human and AI decisions ripple through society, discovering coordination strategies that improve outcomes, and deploying those strategies as live operational tools used by people and institutions. This enables New Zealand to move from reactive responses to anticipatory coordination.

Why Coordination, Not Automation: A central insight motivating this work is that AI’s largest economic and social impact does not come from automating isolated tasks, but from improving coordination across systems. While visible AI adoption in technical roles represents a small fraction of total wage value, coordination and administrative work spans every sector and represents a much larger opportunity. Agentic Zealandia focuses on identifying where human–AI coordination multiplies capability and productivity, while proactively managing the downstream effects on work, services, and wellbeing.

What “A Country of AIs” Means: The phrase “country of AIs” does not mean simulating citizens or replacing people. It refers to a large population of task-level digital agents that represent workflows, roles, and decision points across the economy and public services. These agents coordinate with New Zealand’s 5 million people, supporting real activities in healthcare, emergency response, education, logistics, and environmental management. Humans remain the decision-makers; AI augments coordination, prediction, and execution at scale.

Technology Foundation: The system builds on Large Population Models (LPMs) developed at MIT Project Iceberg, which enable high-resolution simulation of entire populations and their interactions. Iceberg has used this approach to model how AI reshapes work and how ripple effects propagate through economies. PHF Science's ALMA platform applies similar population-scale modeling to health, climate, and biosecurity, and has already been deployed in New Zealand. Agentic Zealandia integrates these foundations to create a seamless pathway from simulation to live coordination, while preserving privacy and national sovereignty.

What We Are Building: Agentic Zealandia consists of three integrated components. First, a human–AI workforce coordination layer that models how AI reshapes work across sectors and supports workforce planning and policy design. Second, a geo-socio-technical integration layer that simulates how decisions in work, health, and the environment interact spatially and socially, and deploys coordination protocols into real systems. Third, a crisis coordination capability that supports large-scale, unplanned events such as pandemics or earthquakes by coordinating human and digital responders in real time.

Why New Zealand: New Zealand's population scale, geographic isolation, and sovereign computing infrastructure make it uniquely suited to deploy coordination systems at a national level. This contained scale is a strategic advantage: it allows rapid iteration, validation, and governance of population-scale human–AI coordination in a real society. By building this capability domestically, New Zealand positions itself as a global leader in responsible AI coordination and resilience.

Why This Matters: Agentic Zealandia turns AI from a collection of tools into national coordination infrastructure. By developing this capability in-country, New Zealand ensures sovereignty over its AI future, increases resilience to systemic shocks, and establishes leadership in how societies integrate AI responsibly. As other nations confront coordination failures in work, health, and supply chains, they will look to models that have already been tested and deployed at national scale.